CPP miniproject

Jan Galek

Michaelmas Term 2025

Question 1

Throughout the question, let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F, G : \mathcal{C} \to \mathcal{D}$ be functors.
(1) Split mono:

- **Naturally split mono implies pointwise split mono.**
  Proof: Suppose $\alpha : F \to G$ is naturally a split mono. Then there exists $\lambda : G \to F$ s.t. $\lambda \circ \alpha = id_F$. Since $\lambda \circ \alpha = id_F$ by definition of vertical composition we immediately get $\lambda_X \circ \alpha_X = (id_F)_X = 1_{FX}$ on components where $X \in Ob(C)$ is arbitrary. Hence λ_X is a left inverse of α_X in $\mathcal{D}$ for arbitrary X. It follows that α_X is a split mono, so α is pointwise split mono. □

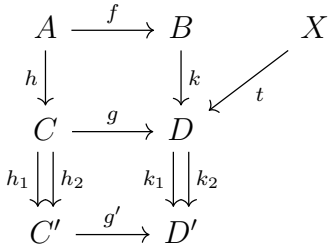
- **Pointwise split mono does not imply naturally split mono.**
  Counterexample:
  Let $G = \{e, a\}$ be a group with the identity e and a satisfying $a^2 = e$. View G as a category with one object $*$ and denote it by $\mathcal{G}$ ($\mathcal{G}(*, *) = G$). Then $\mathbf{Set}^{\mathcal{G}}$ is the same as the category of G-sets, i.e. sets equipped with a specified action of a group G on it. Natural transformations are the same as G-equivariant maps. Let $F, H \in Ob(\mathbf{Set}^{\mathcal{G}})$ be given by $F(*) = \{0, 1\}$ with actions $a \cdot 0 = 1$, $a \cdot 1 = 0$ and $H(*) = \{0, 1, 2\}$ with actions $a \cdot 0 = 1, a \cdot 1 = 0, a \cdot 2 = 2$ (observe that 2 is a fixed point which becomes important later). Define a natural transformation $\alpha : F \to H$ by the inclusion $\alpha_* : F(*) \to H(*)$ given by $\alpha_*(0) = 0, \alpha_*(1) = 1$. Observe that α is pointwise split mono, since α_* has a left inverse in **Set**, e.g. $f : H(*) \to F(*)$ s.t. $f(0) = f(2) = 0, f(1) = 1$.
  However, α is not naturally split mono. Note that it would require a natural transformation $\beta : H \to F$ corresponding to an equivariant map $\beta_* : H(*) \to F(*)$ with $\beta_* \circ \alpha_* = id_F$. Any equivariant map β_* must send fixed points to fixed points. Since 2 is fixed in $H(*)$,

  $$\beta_*(2) = \beta_*(a \cdot 2) = a \cdot \beta_*(2),$$

  so $\beta_*(2)$ must be fixed in $F(*)$. But $F(*)$ has no fixed points, so a contradiction arises. It follows that no satisfying β exists, so α is not a split mono in $\mathbf{Set}^{\mathcal{G}}$. □

(2) Regular mono:

- **Naturally regular mono does not imply pointwise regular mono.**
  Counterexample: Let $\mathcal{C}$ be the category $\mathbf{0} \to \mathbf{1}$. From lectures we know that $\mathcal{D}^{\mathbf{0}\to\mathbf{1}}$ is isomorphic to the arrow category $\mathcal{D}^{\to}$, so a natural transformation α directly corresponds to an arrow in $\mathcal{D}^{\to}$ for any $\mathcal{D}$. To find a counterexample, we need to choose $\mathcal{D}$ s.t. there is $(h,k) \in \mathcal{D}^{\to}(f,g)$ where (h,k) is a regular mono in $\mathcal{D}^{\to}$, but at least one of h,k fails to be a regular mono in $\mathcal{D}$. Define a category $\mathcal{D}$ with the only non-identity arrows as depicted (plus whatever composites are forced) modulo the equations provided below:



$$(1)\ kf = gh,\ (2)\ g'h_1 = k_1 g,\ (3)\ g'h_2 = k_2 g,$$
$$(4)\ h_1 h = h_2 h,\ (5)\ k_1 k = k_2 k,\ (6)\ k_1 t = k_2 t$$

Now, by switching the perspective to $\mathcal{D}^{\to}$, we get the following morphisms: $(h,k) : f \to g$ by (1), $(h_1,k_1) : g \to g'$ by (2), and $(h_2,k_2) : g \to g'$ by (3).
We can quickly see that (h,k) equalises $(h_1,k_1),\ (h_2,k_2)$ since

$$(h_1,k_1) \circ (h,k) = (h_1 h, k_1 k) = (h_2 h, k_2 k) = (h_2,k_2) \circ (h,k) \text{ by (4), (5).}$$

Suppose $(a,b) : q \to g$ is any morphism in $\mathcal{D}^{\to}$ for some $q : Q_1 \to Q_2$ s.t. $(h_1,k_1) \circ (a,b) = (h_2,k_2) \circ (a,b)$. It follows that $ga = bq,\ h_1 a = h_2 a,\ k_1 b = k_2 b$. The only non-identity arrows into C is h. Note that $a \neq 1_C$, since $h_1 \neq h_2$. Hence $a = h$ and $Q_1 = A$.
On the other hand $b : Q_2 \to D$ is forced to satisfy $b \in \{k, g, kf, t\}$. Since $k_1 b = k_2 b$, we get $b \neq g$. Also, we can eliminate t, since $Q_1 = A$ and $b = t$ would render the square (a,b) impossible (as there is no arrow $q : A \to X$). Thus either $b = k$ or $b = kf$.
If $b = k$, then $Q_2 = B$ and $ga = bq$ becomes $gh = kq$. Furthermore, we must have $q = f$, so $(a,b) = (h,k)$, which factors through itself via the identity 1_f.
If $b = kf$, then $Q_2 = A$ and $ga = bq$ becomes $gh = (kf)q$. Hence q is forced to be 1_A. In this case $(a,b) = (h,kf)$ factors through (h,k) via the unique square $(1_A, f) : 1_A \to f$ as we have

$$(h,k) \circ (1_A, f) = (h1_A, kf) = (h,kf).$$

Uniqueness is forced because the only arrow $A \to A$ is 1_A, hence the only arrow $A \to B$ ompatible with a square into f is f itself. This exhaust all posibilities, so (h,k) is the equaliser of (h_1,k_1), (h_2,k_2) in $\mathcal{D}^{\to}$, hence a regular mono.
Importantly, it is not a pointwise regular mono, since k is not a regular mono in $\mathcal{D}$: both k,t equalise k_1,k_2, but t cannot factor through k as there is no arrow $X \to B$. □

- **Pointwise regular mono does not imply naturally regular mono.**
  Counterexample:
  Let $\mathcal{C}$ be a monoid viewed as a category with $Ob(\mathcal{C}) = \{*\}$ and $Ar(\mathcal{C}) = \{1_*, u, u^2, u^3, ...\}$.
  Let $\mathcal{D} = \mathbf{Set_{inj}}$ be the category of sets with morprhisms restricted only to injective functions.
  Subclaim: Every $f \in Ar(\mathbf{Set_{inj}})$ is a regular mono.
  Pf.: Let $f : A \to B \in Ar(\mathbf{Set_{inj}})$ be arbitrary. Define the following two functions $l : B \to B \sqcup B$ and $r : B \to B \sqcup B$ given by $l := \iota_1$ and $r(b) := \begin{cases} \iota_1(b) & \text{if } b \in ran(f) \\ \iota_2(b) & \text{if } b \notin ran(f) \end{cases}$.
  Clearly, l is injective. r is injective as well, since it is injective on $ran(f)$ as well as on $B \setminus ran(f)$. Thus r, s are parallel arrows in $\mathbf{Set_{inj}}$.
  Let $a \in A$. Then $(l \circ f)(a) = \iota_1(f(a))$. Obviously $f(a) \in ran(f)$, so $(r \circ f)(a) = \iota_1(f(a))$. Thus f equalises l, r. Now suppose that $d : D \to B$ is an injective map with $l \circ d = r \circ d$. Let $x \in D$. Then $l(d(x)) = \iota_1(d(x)) = (r \circ d)(x)$. But by definition of r, $r(b) = \iota_1(b)$ iff $b \in ran(f)$, so $d(x) \in ran(f)$. It follows that $ran(d) \subseteq ran(f)$.
  Thanks to that we can define a function $m : D \to A$ given by $m(x) := f^{-1}(d(x))$, where $f^{-1} : ran(f) \to A$ is the inverse of f (exists because f is injective). Then $f \circ m = d$ by construction. For uniqueness, suppose that $m' : D \to A$ satisfies $f \circ m' = d$. Then for each $x \in D$ we have $f(m'(x)) = d(x) = f(m(x))$, and by injectivity of f, we get $m'(x) = m(x)$. Thus $m = m'$, and we can conclude that f is a regular mono. □
  Consider two functors $F, G : \mathcal{C} \to \mathbf{Set_{inj}}$:

  - $F(*) = \mathbb{N}_{\geq 1}$, $F(u) = s' : \mathbb{N}_{\geq 1} \to \mathbb{N}_{\geq 1}$ where $s'(n) = n + 1$
  - $G(*) = \mathbb{N}$, $G(u) = s : \mathbb{N} \to \mathbb{N}$ where $s(n) = n + 1$

  Define $\alpha : F \to G$ by the single component $\alpha_* = i : \mathbb{N}_{\geq 1} \to \mathbb{N}$. Naturality holds immediately due to the equality $s \circ i = i \circ s'$.
  Assume for contradiction that α is a regular mono, i.e. that α is an equaliser of some parallel natural transformations $\beta, \gamma : G \to H$ for some $H : \mathcal{C} \to \mathbf{Set_{inj}}$. Let $H(*) = C$, $H(u) = c : C \to C$. Then $\beta_*, \gamma_* : \mathbb{N} \to C$ are injections satisfying the following equations:

$$c \circ \beta_* = \beta_* \circ s, \quad c \circ \gamma_* = \gamma_* \circ s. \quad (!)$$

  Let $E \subseteq \mathbb{N}$ be the equaliser subset given by $E = \{n \in \mathbb{N} | \beta_*(n) = \gamma_*(n)\}$. Because α is assumed to be an equaliser, we have $E = \mathbb{N}_{\geq 1}$.
  Observe that if $n + 1 \in E$, then $\beta_*(n + 1) = \beta_*(s(n)) = \gamma_*(s(n)) = \gamma_*(n + 1)$. By (!) we get

$$(\beta_* \circ s)(n) = (c \circ \beta_*)(n) = (c \circ \gamma_*)(n) = (\gamma_* \circ s)(n),$$

  so by injectivity of c, $\beta_*(n) = \gamma_*(n)$, so $n \in E$, i.e. E must be closed under predecessors. But $\mathbb{N}_{\geq 1}$ is not closed under predecessors, since $1 \in \mathbb{N}_{\geq 1}$ but $0 \notin \mathbb{N}_{\geq 1}$, a contradiction.
  Therefore α is not a regular mono.

(3) Mono:

- **Naturally mono does not imply pointwise mono.**
  Counterexample: Let $\mathcal{C}$ be the category $\mathbf{0} \to \mathbf{1}$. Define a category $\mathcal{D}$ with the only non-identity arrows as depicted (plus whatever composites are forced) modulo the equation provided below:

$$\begin{array}{ccccc} & & A & & \\ & & \downarrow{\scriptstyle f} & & \\ E & \underset{v}{\overset{u}{\rightrightarrows}} & C & \xrightarrow[b]{} & D \end{array}$$

$$bu = bv \text{ (importantly } u \neq v)$$

  Define functors $F, G : \mathcal{C} \to \mathcal{D}$ by:

  - $F(0) = A, F(1) = C, F(0 \to 1) = f$
  - $G(0) = A, G(1) = D, G(0 \to 1) = bf$

  Now define natural transformation α by components $\alpha_0 = 1_A, \alpha_1 = b$. We have $bf = (bf)1_A$, so naturality holds. Importantly, α is not pointwise mono, since b is not a mono $(u \neq v)$.
  From lectures we know that $\mathcal{D}^{\mathbf{0}\to\mathbf{1}}$ is isomorphic to the arrow category $\mathcal{D}^{\to}$, so α directly corresponds to an arrow in $\mathcal{D}^{\to}$, namely the morphism $(1_A, b)$ from f to bf. $(1_A, b)$ turns out to be a mono. Let $h : X \to Y$ be an arrow in $\mathcal{D}$. Then $\mathcal{D}^{\to}(h, f)$ has at most one element for the following reason: a pair of arrows $(x, y) : h \to f$ must make the diagram below commute:

$$\begin{array}{ccc} X & \xrightarrow{x} & A \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow{y} & C \end{array}$$

  The only arrow into A is 1_A, so $X = A, x = 1_A$. Hence h must be an arrow with domain A, so either $h = 1_A$ or $h = f$. Then the commutativity condition forces y uniquely:

  - if $h = 1_A$, then $y1_A = 1_A f$, so $y = f$
  - if $h = f$, then $f = yf$, so $y = 1_C$ (since the only arrow from C to C is 1_C

  Thus there are no parallel arrows into f in $\mathcal{D}^{\to}$, so every arrow out of f is a mono, particularly $(1_A, b)$. Thus α is a mono in $\mathcal{D}^{\mathcal{C}}$ which concludes the counterexample.

- **Pointwise mono implies naturally mono.**
  Proof: Suppose $\alpha : F \to G$ is a pointwise mono, i.e. for all $X \in Ob(\mathcal{C})$ α_X is a mono in $\mathcal{D}$. Given natural transformations $\mu, \eta : H \to F$ s.t. $\alpha \circ \mu = \alpha \circ \eta$, by definition of vertical composition we get $(\alpha \circ \mu)_X = \alpha_X \circ \mu_X = \alpha_X \circ \eta_X = (\alpha \circ \eta)_X$. By monicity of α_X, $\mu_X = \eta_X$. Since X was arbitrary, it follows that $\mu = \eta$. □

Question 2

Observe that if Σ is empty, then the category $\mathbf{Alg}_\varnothing$ is isomorphic to $\mathbf{Set}$, since every function vacuously satisfies the definition of Σ-homomorphism. Formally, consider the functors $F : \mathbf{Alg}_\varnothing \to \mathbf{Set}$ given by $F((A, \varnothing)) = A$ on objects and $F(h) = h$ on arrows and $G : \mathbf{Set} \to \mathbf{Alg}_\varnothing$ given by $G(A) = (A, \varnothing)$ and $G(h) = h$. Furthermore, if Σ is empty, for every non-empty set A, every element $a \in A$ is (again vacuously) a kixpoint. Hence $\mathbf{KAlg}_\varnothing \cong \mathbf{Set}_{\neq\varnothing}$, the category of all non-empty sets and functions.
For the same reason $\mathbf{XAlg}_\varnothing \cong \mathbf{Set}_{\mathbf{pt}}$, the category of pointed sets. Therefore, throughout the rest of the question I assume that Σ is non-empty, unless explicitly stated otherwise.
(a) Initial object:

- $\mathbf{Alg}_{\mathbf{\Sigma}}$:
  In this category initial objects exist for all signatures Σ, but we need to differentiate between two cases.
  If $ar : \Sigma \to \mathbb{N}$ is s.t. $\forall f \in \Sigma. ar(f) \neq 0$, then the initial object is simply a Σ-algebra with empty domain. Since morphisms in $\mathbf{Alg}_{\mathbf{\Sigma}}$ are by definition functions of the form $h : A \to B$, if $A = \varnothing$, then for all $B \in Ob(\mathbf{Alg}_{\mathbf{\Sigma}})$ there exists a unique $h : \varnothing \to B$, which vacously satisfies homomorphism condition.
  If $ar : \Sigma \to \mathbb{N}$ is s.t. $\exists c \in \Sigma. ar(c) = 0$, then there is no Σ-algebra $\mathcal{A}$ with $A = \varnothing$ as $c^{\mathcal{A}} : A^0 \to A$ cannot exist. Let CT_Σ be a set of closed Σ-terms (i.e. without using any variables) built in the following way:

  - if $c \in \Sigma$ is s.t. $ar(c) = 0$, then $c \in CT_\Sigma$
  - if $f \in \Sigma$ and $ar(f) = n$, then $\forall t_1, ..., t_n \in CT_\Sigma. f(t_1, ..., t_n) \in CT_\Sigma$

  Now, define $\mathcal{T} = (CT_\Sigma, (f^{\mathcal{T}})_{f\in\Sigma})$, where for each 0-ary $c \in \Sigma$, $c^{\mathcal{T}} : \{*\} \to CT_\Sigma$ is the unique function with $c^{\mathcal{T}}(*) = c$ (informally identified with the symbol c), and for each $f \in \Sigma$ with $ar(f) = n > 0$ we have
  $f^{\mathcal{T}} : CT_\Sigma^n \to CT_\Sigma$ given by term formation $f^{\mathcal{T}}(t_1, ..., t_n) := f(t_1, .., t_n)$. Given a Σ-algebra $\mathcal{A}$ there exists a unique morphism $eval_A : CT_\Sigma \to A$ defined by $eval_{\mathcal{A}}(c) = c^{\mathcal{A}}$ for $c \in \Sigma$ with $ar(c) = 0$ and $eval_{\mathcal{A}}(f^{\mathcal{T}}(t_1, ..., t_n)) = f^{\mathcal{A}}(eval_{\mathcal{A}}(t_1), ..., eval_{\mathcal{A}}(t_n))$ for $f \in \Sigma$ with $ar(f) = n > 0$. Uniqueness follows from the following induction on term complexity:
  Let $h : CT_\Sigma \to A$ be a Σ-homomorphism from $\mathcal{T}$ to $\mathcal{A}$.

  - if $t = c$, $ar(c) = 0$, then $h(t) = h(c^{\mathcal{T}}) = c^{\mathcal{A}} = eval_{\mathcal{A}}(c)$.
  - Suppose $t = f(t_1, ..., t_n)$ and that $h(t') = eval_{\mathcal{A}}(t')$ for all terms t' with complexity lower than t (IH). Then by the homomorphism property $h(t) = h(f^{\mathcal{T}}(t_1, ..., t_n)) = f^{\mathcal{A}}(h(t_1), ..., h(t_n)) = f^{\mathcal{A}}(eval_{\mathcal{A}}(t_1), ..., eval_{\mathcal{A}}(t_n)) = eval_{\mathcal{A}}(f^{\mathcal{T}}(t_1, ..., t_n)) = eval_{\mathcal{A}}(t)$

  Thus $\mathcal{T}$ is an initial object.

- $\mathbf{KAlg}_{\mathbf{\Sigma}}$:
  In this case there exists a specific example without an initial object, namely $\mathbf{KAlg}_\varnothing \cong \mathbf{Set}_{\neq\varnothing}$. Suppose for contradiction that $\mathbf{Set}_{\neq\varnothing}$ has initial object I. Since empty set is not present among the objects of this category $I \neq \varnothing$. Consider the set $\{0, 1\}$. Then there exist $f : I \to \{0, 1\}$ s.t. for $x \in I$ $f(x) = 0$ and $g : I \to \{0, 1\}$ s.t. for $x \in I$ $g(x) = 1$, contradicting initiality of I.
  Thus $\mathbf{KAlg}_\varnothing \cong \mathbf{Set}_{\neq\varnothing}$ does not have initial objects.

- $\mathbf{XAlg}_{\mathbf{\Sigma}}$:
  Initial objects exist. Consider pairs of the form $(\mathcal{S}, *)$, where $\mathcal{S}$ is a Σ-algebra with

singleton domain $S = \{*\}$. Since every function into S is constant, $*$ is a kixpoint, so $(\mathcal{S}, *) \in Ob(\mathbf{XAlg_\Sigma})$. Then for arbitrary object $(\mathcal{A}, a_0)$ there exists only one point-preserving homomorphism $h : S \to A$, namely the one given by $h(*) = a_0$.

(b) Terminal objects:

- $\mathbf{Alg_\Sigma}$:
  In this category terminal objects exist for all Σs and are simply Σ-algebras with singleton domains. Suppose $\mathcal{S}$ is an object with domain $S = \{*\}$. Then for arbitrary $\mathcal{A}$ there exists a unique (since codomain is a singleton) constant function $h : A \to S$ which trivially satisfies the homomorphism definition.

- $\mathbf{KAlg_\Sigma}$:
  Since all Σ-algebras with a singleton domain have a kixpoint (the only element of a domain), they are included in $\mathbf{KAlg_\Sigma}$ and also serve as terminal objects in this category.

- $\mathbf{XAlg_\Sigma}$:
  Terminal objects exist. By analogy with $\mathbf{Set_{pt}}$ pairs of the form $(\mathcal{S}, *)$, where $\mathcal{S}$ is a Σ-algebra with singleton domain $S = \{*\}$ are terminal objects, since for arbitrary object $(\mathcal{A}, a_0)$ there exists only one point-preserving homomorphism, i.e. the unique function $h : A \to S$.

(c) (binary) products:

- $\mathbf{Alg_\Sigma}$:
  Given two arbitrary Σ-algebras $\mathcal{A}$ and $\mathcal{B}$, their product $\mathcal{A} \times \mathcal{B}$ is a Σ-algebra with domain $A \times B$ and interpretation for every function symbol $f \in \Sigma$ with $ar(f) = n$ is given by

$$f^{\mathcal{A}\times\mathcal{B}}((a_1, b_1), ..., (a_n, b_n)) := (f^{\mathcal{A}}(a_1, ..., a_n), f^{\mathcal{B}}(b_1, ..., b_n))$$

  and $\pi_1 : A \times B \to A$, $\pi_2 : A \times B \to B$ are the same as in **Set**. Observe that π_1, π_2 are indeed arrows in $\mathbf{Alg_\Sigma}$, since

$$\pi_1(f^{\mathcal{A}\times\mathcal{B}}((a_1, b_1), ..., (a_n, b_n))) = f^{\mathcal{A}}(a_1, ..., a_n) = f^{\mathcal{A}}(\pi_1((a_1, b_1)), ..., \pi_1((a_n, b_n)))$$

$$\pi_2(f^{\mathcal{A}\times\mathcal{B}}((a_1, b_1), ..., (a_n, b_n))) = f^{\mathcal{B}}(b_1, ..., b_n) = f^{\mathcal{B}}(\pi_2((a_1, b_1)), ..., \pi_2((a_n, b_n)))$$

  To check that product conditions hold, let $g : \mathcal{C} \to \mathcal{A}$ and $h : \mathcal{C} \to \mathcal{B}$ for some arbitrary $\mathcal{C}$. Then as in **Set** there exists a unique arrow $m := \langle g, h\rangle : \mathcal{C} \to \mathcal{A} \times \mathcal{B}$ given by $m(c) = (g(c), h(c))$, so $\pi_1 \circ m = g$ and $\pi_2 \circ m = h$. As before, observe that m is a Σ-homomorphism:

$$m(f^{\mathcal{C}}(c_1, ..., c_n)) = (g(f^{\mathcal{C}}(c_1, ..., c_n), h(f^{\mathcal{C}}(c_1, ..., c_n))) =$$

$$= (f^{\mathcal{A}}(g(c_1), ..., g(c_n)), f^{\mathcal{B}}(h(c_1), ..., h(c_n))) = f^{\mathcal{A}\times\mathcal{B}}(m(c_1), ..., m(c_n))$$

  Uniqueness of m follows from the uniqueness of underlying set map into $A \times B$.

- $\mathbf{KAlg_\Sigma}$:
  Lemma: If $\mathcal{A}, \mathcal{B}$ have kixpoints, then $\mathcal{A} \times \mathcal{B}$ have a kixpoint as well.
  Proof: Let k, l be respectively kixpoints of $\mathcal{A}, \mathcal{B}$. Fix $f \in \Sigma$. If $f^{\mathcal{A}\times\mathcal{B}}((k, l), (k, l), ..., (k, l))) = (k, l)$, then $f^{\mathcal{A}}, f^{\mathcal{B}}$ are forced to be constant functions returning values k and l respectively. It follows that $f^{\mathcal{A}\times\mathcal{B}}$ is a constant function itself, so (k, l) is a kixpoint. □

  Hence $\mathbf{KAlg_\Sigma}$ inherits products from $\mathbf{Alg_\Sigma}$.

- **$\mathbf{XAlg_\Sigma}$**:
  Products exist. For objects $(\mathcal{A}, k), (\mathcal{B}, l)$ their product is just $(\mathcal{A} \times \mathcal{B}, (k, l))$. As it was shown before, (k, l) is a kixpoint. The projections preserve the distinguished element, and the mediating map from any $(\mathcal{C}, m)$ is the usual product homomorphism, which also preserves the point because its components do.

(d) (binary) coproducts:
The following definition will be useful in subsequent constructions (from [1], p.35 def. 5.1.):
A congruence on $\mathcal{A}$ is an equivalence relation $\equiv$ on A s.t. for every $f \in \Sigma$ of arity n,

$$a_i \equiv a_i' \text{ for all } i = 1, ..., n \implies f^{\mathcal{A}}(a_1, ..., a_n) \equiv f^{\mathcal{A}}(a_1', ..., a_n')$$

- **$\mathbf{Alg_\Sigma}$**: Coproducts exist. Given two Σ-algebras $\mathcal{A}$ and $\mathcal{B}$, their coproduct $\mathcal{A} + \mathcal{B}$ can be constructed in the following way: take $A \sqcup B$ and form a set $T_\Sigma(A \sqcup B)$ consisting of Σ-terms with variables from $A \sqcup B$. Let $\iota_1 : A \to A \sqcup B$, $\iota_2 : B \to A \sqcup B$ be the standard set-inclusions. Also, let $\tau : A \sqcup B \to T_\Sigma(A \sqcup B)$ be the inclusion of elements of $A \sqcup B$ in $T_\Sigma(A \sqcup B)$ as variable terms, and define $j_1 = \tau \circ \iota_1$, $j_2 = \tau \circ \iota_2$.
  Now we need the smallest congruence $\sim$ on $T_\Sigma(A \sqcup B)$ s.t. for each $f \in \Sigma$ with $ar(f) = n$ and each $a_1, ..., a_n \in A$, $j_1(f^{\mathcal{A}}(a_1, ..., a_n)) \sim f^{\mathcal{T}_\Sigma(A \sqcup B)}(j_1(a_1), ..., j_1(a_n))$ and similarly for each $b_1, ..., b_n \in B$ $j_2(f^{\mathcal{B}}(b_1, ..., b_n)) \sim f^{\mathcal{T}_\Sigma(A \sqcup B)}(j_2(b_1), ..., j_2(b_n))$. Let R the set of pairs used to generate $\sim$, i.e. for each $f \in \Sigma$ with $ar(f) = n$ and $a_1, ..., a_n \in A$,

  $$(j_1(f^{\mathcal{A}}(a_1, ..., a_n)), f^{\mathcal{T}(A \sqcup B)}(j_1(a_1), ..., j_1(a_n))) \in R,$$

  and the same for B. Then we can define

  $$\Theta := \{\theta \subseteq T_\Sigma(A \sqcup B) \times T_\Sigma(A \sqcup B) | \theta \text{ is a congruence and } R \subseteq \theta\} \text{ and } \sim := \bigcap_{\theta \in \Theta} \theta.$$

  Note that everything works:

  - $\Theta \neq \varnothing$ since $T_\Sigma(A \sqcup B) \times T_\Sigma(A \sqcup B) \in \Theta$
  - $\sim$ is a congruence, since congruences are exactly equivalence relations closed under all basic operations, and each of those closure properties is preserved under arbitrary intersections
  - by construction $R \subseteq \sim$
  - if $\theta \in \Theta$, then by construction $\sim \subseteq \theta$

  Now, let $Q = T_\Sigma(A \sqcup B)/\sim$ and $q : T_\Sigma(A \sqcup B) \to Q$ be given by $q(t) = [t]_\sim$.
  $\mathcal{Q} = (Q, (f^{\mathcal{Q}})_{f \in \Sigma})$, where $f^{\mathcal{Q}}([t_1], ..., [t_n]) = [f(t_1, .., t_n)]$, is the desired coproduct along with arrows $y_1 = q \circ j_1$ and $y_2 = q \circ j_2$.
  To check that the universal property holds, let $\mathcal{C}$ be a Σ-algebra with homomorphisms $h : \mathcal{A} \to \mathcal{C}$, and $g : \mathcal{B} \to \mathcal{C}$. Define a function $s : A \sqcup B \to C$ by

  $$s(\iota_1(a)) := h(a), \ s(\iota_2(b)) := g(b).$$

  At this point I will alow myself to state without a proof that $T_\Sigma(A \sqcup B)$ is free over $A \sqcup B$. The proof of this fact is provided in $(g)(iii)$. By freenes, s extends to a unique Σ-homomorphism $s^\# : \mathcal{T}(A \sqcup B) \to \mathcal{C}$.
  Now observe that if $(u, v) \in R$, then $s^\#(u) = s^\#(v)$:
  we know that $s^\#$ is a homomorphism and $s^\# \circ j_1 = h$, so

  $$s^\#(f^{\mathcal{T}(A \sqcup B)}(j_1(a_1), ..., j_1(a_n))) = f^{\mathcal{C}}(s^\#(j_1(a_1)), ..., s^\#(j_1(a_n))) = f^{\mathcal{C}}(h(a_1), ..., h(a_n))$$

$$= h(f^{\mathcal{A}}(a_1, ..., a_n)) = s^{\#}(j_1(f^{\mathcal{A}}(a_1, ..., a_n)))$$

The same holds for B. Thus $s^{\#}$ equalises every pair in R.
Let $K = \{(x, y) \in T_\Sigma(A \sqcup B) \times T_\Sigma(A \sqcup B) | s^{\#}(x) = s^{\#}(y)\}$. K forms an equivalence relation. Take $f \in \Sigma$ of arity n and assume that $(x_i, y_i) \in K$ for each $i = 1, ..., n$. Then

$$s^{\#}(f^{\mathcal{T}(A \sqcup B)}(x_1, ..., x_n)) = f^{\mathcal{C}}(s^{\#}(x_1), ..., s^{\#}(x_n))$$

$$= f^{\mathcal{C}}(s^{\#}(y_1), .., s^{\#}(y_n)) = s^{\#}(f^{\mathcal{T}(A \sqcup B)}(y_1, ..., y_n)),$$

so K is a congruence. It follows that $\sim \subseteq K$, so if $t \sim t'$, then $s^{\#}(t) = s^{\#}(t')$.
Finally, we can define a homomorphism $s_! : \mathcal{Q} \to \mathcal{C}$ by $s_!([t]) := s^{\#}(t)$. This is well-defined by previous steps, and by construction $s_! \circ q = s^{\#}$. To see that $s_!$ is a homomorphism, for n-ary $f \in \Sigma$

$$s_!(f^{\mathcal{Q}}([t_1], ..., [t_n]) = s_!([f(t_1, ..., t_n)]) = s^{\#}(f(t_1, ..., t_n))$$

$$= f^{\mathcal{C}}(s^{\#}(t_1), ..., s^{\#}(t_n)) = f^{\mathcal{C}}(s_!([t_1]), ..., s_!([t_n])).$$

Now for $a \in A$,

$$s_!(y_1(a)) = s_!(q(j_1(a))) = s^{\#}(j_1(a)) = s(\iota_1(a)) = h(a),$$

and similarly for $b \in B$, $s_! \circ y_2 = g$. Hence $s_!$ is candidate for the unique mediating map. Uniqueness holds, if $u : \mathcal{Q} \to \mathcal{C}$ is a homomorphism with $u \circ y_1 = h$, $u \circ y_2 = g$, then $u \circ q : \mathcal{T}(A \sqcup B) \to \mathcal{C}$ is a homomorphism, and it agrees on variables from $A \sqcup B$ with s, hence by freenes of $\mathcal{T}(A \sqcup B)$ we must have $u \circ q = s_! \circ q = s^{\#}$. Since q is a quotient map, it is surjective so $u = s_!$.
This concludes the construction of $\mathcal{Q}$ with y_1, y_2 as coproduct.

- $\mathbf{KAlg_\Sigma}$:
  For $\Sigma = \{u\}$ with $ar(u) = 1$ $\mathbf{KAlg_\Sigma}$ does not have coproducts. Observe that for such Σ, k is a kixpoint for $\mathcal{A}$ if and only if $s^{\mathcal{A}}(k) \neq k$ or $\forall x \in A. s^{\mathcal{A}}(x) = k$.
  Consider $\mathcal{S} = (\{*\}, (u^{\mathcal{S}}))$ with the only possible $u^{\mathcal{S}} : \{*\} \to \{*\}$ given by $u^{\mathcal{S}}(*) = *$. It follows that $*$ is a kixpoint. Suppose for contradiction that there exists a coproduct $\mathcal{P} = \mathcal{S} + \mathcal{S}$ along with $\iota_1 : \mathcal{S} \to \mathcal{P}$ and $\iota_2 : \mathcal{S} \to \mathcal{P}$. Let $a := \iota_1(*)$ and $b := \iota_2(*)$. Since ι_1, ι_2 are homomorphisms and $u^{\mathcal{S}}(*) = *$, we have $u^{\mathcal{P}}(a) = a$ and $u^{\mathcal{P}}(b) = b$, so a, b are fixed points of $u^{\mathcal{P}}$. Consider $\mathcal{C}$ with $C = \{x, y, z\}$ with $u^{\mathcal{C}}(x) = x$, $u^{\mathcal{C}}(y) = y$ and $u^{\mathcal{C}}(z) = x$. Then z is a kixpoint, so $\mathcal{C} \in \mathbf{KAlg_\Sigma}$. There are homomorphisms f, g from $\mathcal{S}$ to $\mathcal{C}$ given by $f(*) = x$ and $g(*) = y$. By the coproduct property there must exist a unique m from $\mathcal{P}$ to $\mathcal{C}$ s.t. $m(a) = x$ and $m(b) = y$. Thus $a \neq b$.
  By assumption $\mathcal{P} \in \mathbf{KAlg_\Sigma}$, so $\mathcal{P}$ has some kixpoint l. Observe that $l \neq a, b$, because a, b are fixed points, whereas $u^{\mathcal{P}}$ is not constant, so they cannot be kixpoints. Furthermore, $s^{\mathcal{P}}(l) \neq l$. Take two disjoint copies of the domain of $\mathcal{P}$, P_1 and P_2. Define an equivalence relation $\sim$ by identifying $a_1 \sim a_2$ and $b_1 \sim b_2$ and nothing else. Let Q be the quotient set. Define a Σ-algebra $\mathcal{Q}$ with domain Q by $u^{\mathcal{Q}}(x) = [u^{\mathcal{P}}(x)]_\sim$. This is well-defined because the only identifications were $a_1 \sim a_2$, $b_1 \sim b_2$, and both a, b are fixed by $u^{\mathcal{P}}$. The equivalence classes of l_1, l_2 are kixpoints in $\mathcal{Q}$ since $u^{\mathcal{P}}(l) \neq l$ and $l \neq a, b$. Thus $\mathcal{Q} \in \mathbf{KAlg_\Sigma}$. Now there are two homomorphisms: $e_1 : P \to Q$ and $e_2 : P \to Q$ given by inclusion into P_1 and P_2 respectively. They satisfy $e_1(a) = e_2(a)$ and $e_1(b) = e_2(b)$ because of the previous identification. However $e_1(l) \neq e_2(l)$ since it is not the case that $l_1 \sim l_2$. Hence e_1, e_2 are distinct homomorphisms from $\mathcal{P}$ to $\mathcal{Q}$ extending the same pair of maps $\{*\} \to Q$. That contradicts the required uniqueness condition. Therefore $\mathbf{KAlg_\Sigma}$ does not have coproducts for every Σ.

- $\mathbf{XAlg_\Sigma}$:
  This category has coproducts. At this point, it is worth to consider an additional category $\mathbf{Alg_\Sigma^{pt}}$ consisting of pairs $(\mathcal{A}, a_0)$ where $\mathcal{A}$ is a Σ-algebra and $a_0 \in A$ is a distinguished element. $\mathbf{XAlg_\Sigma}$ is a (full) subcategory of $\mathbf{Alg_\Sigma^{pt}}$ where one keeps only pairs where distinguished point is a kixpoint. Importantly, $\mathbf{XAlg_\Sigma} \cong \mathbf{K} \subseteq \mathbf{Alg_{\Sigma_+}}$ where $\mathbf{Alg_{\Sigma_+}}$ is just a regular category of all algebras over $\Sigma_+ = \Sigma \cup \{c\}$ with additional 0-ary function symbol c corresponding to a distinguished element. An object $(\mathcal{A}, k)$ of $\mathbf{XAlg_\Sigma}$ corresponds to an object in $\mathbf{K}$, a Σ_+-algebra with $c^\mathcal{A} = k$ satisfying for each $f \in \Sigma$ of arity n the kixpoint condition
  $$f^\mathcal{A}(c^\mathcal{A}, ..., c^\mathcal{A}) = c^\mathcal{A} \implies \forall a_1, ..., a_n. f^\mathcal{A}(a_1, ..., a_n) = c^\mathcal{A}$$
  Now the aim is to show that the inclusion $\mathbf{K} \to \mathbf{Alg_{\Sigma_+}}$ has a left adjoint, since $\mathbf{Alg_{\Sigma_+}}$ have coproducts and left adjoints preserve colimits. For any Σ_+-algebra $\mathcal{A}$, define
  $$\Theta := \{\theta | \theta \text{ is a congruence on } A \text{ and } \mathcal{A}/\theta \in Ob(\mathbf{K})\}, \quad \theta_A := \bigcap_{\theta \in \Theta} \theta.$$
  Again, $\Theta \neq \varnothing$, since $A \times A \in \Theta$ (it yields a quotient algebra with just 1 element in its domain). The more involved part is to show that $\mathcal{A}/\theta_A \in Ob(\mathbf{K})$. Fix $f \in \Sigma$ with arity n, and suppose that in $\mathcal{A}/\theta_A$ $f^{\mathcal{A}/\theta_A}([c^\mathcal{A}], ..., [c^\mathcal{A}]) = [c^\mathcal{A}]$. We get a chain of equivalences:
  $f^{\mathcal{A}/\theta_A}([c^\mathcal{A}], ..., [c^\mathcal{A}]) = [c^\mathcal{A}] \iff (f^\mathcal{A}(c^\mathcal{A}, ..., c^\mathcal{A}), c^\mathcal{A}) \in \theta_A \iff$
  $\forall \theta \in \Theta,\ (f^\mathcal{A}(c^\mathcal{A}, ..., c^\mathcal{A}), c^\mathcal{A}) \in \theta \iff \forall \theta \in \Theta,\ f^{\mathcal{A}/\theta}([c^\mathcal{A}]_\theta, ..., [c^\mathcal{A}]_\theta) = [c^\mathcal{A}]_\theta$.
  But $\mathcal{A}/\theta \in Ob(\mathbf{K})$, so for each $\theta \in \Theta$ and all $a_1, ..., a_n \in A$, $(f^\mathcal{A}(a_1, ..., a_n), c^\mathcal{A}) \in \theta$. Hence it holds in the intersection θ_A, which is exactly $f^{\mathcal{A}/\theta_A}([a_1]_{\theta_A}, ..., [a_n]_{\theta_A}) = [c^\mathcal{A}]_{\theta_A}$.
  Thus $\mathcal{A}/\theta_A \in Ob(\mathbf{K})$ and we have the quotient map $q : A \to A/\theta_A$ given by $q(a) = [a]$.
  Let $h : \mathcal{A} \to \mathcal{B}$ be a Σ_+-homomorphism with $\mathcal{B} \in Ob(\mathbf{K})$.
  Then $\chi = \{(x, y) \in A \times A | h(x) = h(y)\}$ is a congruence and by the First Isomorphism Theorem $\mathcal{A}/\chi \cong ran(h)$ (from [1], p.46, thm. 6.12). Since h is a homomorphism, $ran(h)$ forms a subalgebra of $\mathcal{B}$, and since $h(c^\mathcal{A}) = c^\mathcal{B} \in ran(h)$ it is an object in $\mathbf{K}$. Thus $\mathcal{A}/\chi$ is an object as well, and $\chi \in \Theta$, so $\theta_A \subseteq \chi$. Hence if $(a, a') \in \theta_A$,then $h(a) = h(a')$, so $h^\#([a]) := h(a)$ is well-defined and is a homomorphism. Let $q : \mathcal{A} \to \mathcal{A}/\theta_A$ be a quotient map. Then
  $$h = h^\# \circ q,$$
  and uniqueness follows since q is surjective.
  Therefore $L : \mathbf{Alg_{\Sigma_+}} \to \mathbf{K}$ given by $L(\mathcal{A}) = \mathcal{A}/\theta_A$ satisfies $L \vdash I$.

(e) equalisers:

- $\mathbf{Alg_\Sigma}$: Equalisers exist. Let $g, h : A \to B$ be parallel Σ-homomorphisms from $\mathcal{A}$ to $\mathcal{B}$. Define the underlying set $E = \{a \in A | g(a) = h(a)\} \subseteq A$. Let $e : E \to A$ be the inclusion. The set E will serve as a domain of Σ-algebra $\mathcal{E}$. For each $f \in \Sigma$ with $ar(f) = n$, define $f^\mathcal{E} : E^n \to E$ by restriction $f^\mathcal{E}(a_1, ..., a_n) := f^\mathcal{A}(a_1, ..., a_n)$ provided $f^\mathcal{A}(a_1, ..., a_n) \in E$. It follows that we must check closure.
  If $n = 0$, then $f^\mathcal{A} \in A$ satisfies $g(f^\mathcal{A}) = f^\mathcal{B} = h(f^\mathcal{A})$, so $f^\mathcal{A} \in E$.
  If $n > 0$, let $a_1, ...a_n \in E$. Then by definition of E, $g(a_i) = h(a_i)$ for all $i = 1, ..., n$. Since g, h are homomorphisms, we have
  $$g(f^\mathcal{A}(a_1, ..., a_n)) = f^\mathcal{B}(g(a_1), ..., g(a_n)) = f^\mathcal{B}(h(a_1), ..., h(a_n)) = h(f^\mathcal{A}(a_1, ..., a_n))$$
  Hence $f^\mathcal{A}(a_1, ..., a_n) \in E$, so the restriction works as expected and $\mathcal{E} = (E, (f^\mathcal{E})_{f \in \Sigma})$ is a Σ-algebra. By definition of E, e equalises g and h, since for all $x \in E$, we have $g(e(x)) = h(e(x))$, so $g \circ e = h \circ e$. Now, let $d : D \to A$ be any homomorphism from $\mathcal{D}$ to

$\mathcal{A}$ satisfying $g \circ d = h \circ d$. Then for every $y \in D$, $g(d(y)) = h(d(y))$, so $d(y) \in E$. Hence there is a unique function $m : D \to E$ s.t. $d = e \circ m$.
It remains to show that m is a Σ-homomorphism. Let $f \in \Sigma$ have arity n. Then

$$e(m(f^{\mathcal{D}}(y_1, ..., y_n))) = d(f^{\mathcal{D}}(y_1, ..., y_n)) = f^{\mathcal{A}}(d(y_1), ..., d(y_n)) = e(f^{\mathcal{E}}(m(y_1), ..., m(y_n)))$$

Since e is an inclusion, it is injective, so $m(f^{\mathcal{D}}(y_1, ..., y_n)) = f^{\mathcal{E}}(m(y_1), ..., m(y_n))$. This concludes the construction of $e : E \to A$ and $\mathcal{E}$ as equaliser of g, h.

- $\mathbf{KAlg_\Sigma}$:
For $\Sigma = \{u\}$ with $ar(u) = 1$ the category does not have equalisers.
Let $\mathcal{A}$ be a Σ-algebra s.t. $A = \{0, 1, 2\}$ and $s^{\mathcal{A}}(0) = 0, s^{\mathcal{A}}(1) = 1, s^{\mathcal{A}}(2) = 1$. It follows that 2 is a kixpoint, so $\mathcal{A} \in \mathbf{KAlg_\Sigma}$.
Let $\mathcal{B}$ be a Σ-algebra s.t. $B = \{0, 1\}$ and $s^{\mathcal{B}}(0) = 1,\ s^{\mathcal{B}}(1) = 1$. Then 1 is a kixpoint, so $\mathcal{B} \in \mathbf{KAlg_\Sigma}$.
Now, define homomorphisms $f, g : A \to B$ in $\mathbf{KAlg_\Sigma}$ by

$$g(0) = g(1) = g(2) = 1 \quad f(0) = f(1) = 1,\ f(2) = 0$$

These are homomorphisms since for $a = 0, 1$ we must have $h(a) = s^{\mathcal{B}}(h(a))$ which in turn forces $h(0) = h(1) = 1$, and for $a = 2$ we need $h(1) = s^{\mathcal{B}}(h(2))$, which holds for both choices $h(2) = 0, 1$.
In $\mathbf{Alg_\Sigma}$, the equaliser of f, g is the subalgebra $E = \{a \in A | f(a) = g(a)\} = \{0, 1\}$ with $s^{\mathcal{E}}$, the identity on E. But E has no kixpoints. Thus the equaliser in $\mathbf{Alg_\Sigma}$ is not an object of $\mathbf{KAlg_\Sigma}$.
Now suppose for contradiction that there is an equaliser $e : E' \to A$ of f, g in $\mathbf{KAlg_\Sigma}$. From lectures we know that e is mono. To derive a contradiction we need additional fact, namely that every mono in $\mathbf{KAlg_\Sigma}$ is an injective function:
Let $\mathcal{T}_x = (T_\Sigma(\{x\}), (s^{\mathcal{T}_x}))$ be where $T_\Sigma(\{x\}) = \{x, s(x), s^2(x), ...\}$ and $s^{\mathcal{T}_x}(s^n(x)) = s^{n+1}(x)$. Observe that $\mathcal{T}_x \in \mathbf{KAlg_\Sigma}$, since for any $k = s^n(x)$, $s^{\mathcal{T}_x}(k) = s^{\mathcal{T}_x}(s^n(x)) = s^{n+1}(x) \neq k$. For any $\mathcal{E}'$ with domain E' and any element $a \in E'$, there is a unique homomorphism $\eta_a : T_\Sigma(\{x\}) \to E'$ with $\eta_a(x) = a,\ \eta_a(s^n(x)) = (s^{\mathcal{E}'})^n(a)$.
Assume that e is not injective, so there are $a, b \in E'$ s.t. $e(a) = e(b)$ and $a \neq b$. Consider η_a, η_b. Compute the following on x:

$$(e \circ \eta_a)(x) = e(a), \quad (e \circ \eta_b)(x) = e(b)$$

Since $e(a) = e(b)$, we have $(e \circ \eta_a)(x) = (e \circ \eta_b)(x)$. But x generates $T_\Sigma(\{x\})$, so equality on the generator forces equality everywhere. Hence $e \circ \eta_a = e \circ \eta_b$, and by monicity of e we get $\eta_a = \eta_b$. Evaluating at x gives $a = b$, contradicion. Thus e must be injective.
It follows that $\mathcal{E}'$ embeds in $\mathcal{A}$, and because $f \circ e = g \circ e$, its image must lie in $E = \{0, 1\}$. So $im(e)$ is a subalgebra of $\mathcal{E}$, hence the only options are either $\{0\}, \{1\}$ or $\{0, 1\}$.
Note that it cannot be a singleton: there is a homomorphism h from the algebra $\mathcal{T}_{x,y} = (T_\Sigma(\{x, y\}), (s^{\mathcal{T}_{x,y}}))$ to $\mathcal{A}$ sending the two generators x, y to $0, 1$ and this h equalises f, g since f, g agree on $\{0, 1\}$. However, h cannot factor through a singleton.
On the other hand, if $im(e) = \{0, 1\}$, then $\mathcal{E}' \cong \mathcal{E}$, contradicting that $\mathcal{E}'$ must have a kixpoint.
Therefore the equaliser does not exist in $\mathbf{KAlg_\Sigma}$.

- $\mathbf{XAlg_\Sigma}$:
Equalisers exist for every Σ. Let Σ be arbitrary. Take parallel arrows in $\mathbf{XAlg_\Sigma}$, $h, g : (\mathcal{A}, k) \to (\mathcal{B}, l)$, so $h, g : A \to B$ are Σ-homomorphisms with $h(k) = g(k) = l$. Let

$E := \{a \in A | h(a) = g(a)\} \subseteq A$. Observe that $\mathcal{E}$ with domain E and interpretations of function symbols inherited from $\mathcal{A}$ is a subalgebra of $\mathcal{A}$: if $f \in \Sigma$ has arity n and $a_1, ..., a_n \in E$, then

$$h(f^{\mathcal{A}}(a_1, ..., a_n)) = f^{\mathcal{B}}(h(a_1), ..., h(a_n)) = f^{\mathcal{B}}(g(a_1), ..., g(a_n)) = g(f^{\mathcal{A}}(a_1, ..., a_n)),$$

so $f^{\mathcal{A}}(a_1, ..., a_n) \in E$. Hence the inclusion $e : E \to A$ is a homomorphism, and $h \circ e = g \circ e$. Furthermore $k \in E$ since $h(k) = g(k)$. Hence $(\mathcal{E}, k)$ is a pointed Σ-algebra. However, it still remains to check if k is a kixpoint in $\mathcal{E}$. Fix $f \in \Sigma$:

- If $f^{\mathcal{A}}(k, ..., k) \neq k$, then $f^{\mathcal{E}}(k, ..., k) = f^{\mathcal{A}}(k, ..., k) \neq k$.
- If $f^{\mathcal{A}}(k, ..., k) = k$, then $f^{\mathcal{A}}$ must be constant with value k for all elements of A^n. Hence the restriction $f^{\mathcal{E}}$ to E^n must be constant with value k as well.

Therefore k is a kixpoint in $\mathcal{E}$, so $(\mathcal{E}, k)$ is an object in $\mathbf{XAlg_\Sigma}$.
Finally, given any $d : (\mathcal{C}, m) \to (\mathcal{A}, k)$ in $\mathbf{XAlg_\Sigma}$ with $h \circ d = g \circ d$, we have $d(c) \in E$ for every $c \in C$, so d factors uniquely through the inclusion e as a Σ-homomorphism $q : C \to E$. Since $d(m) = k$ and e is inclusion, we have $q(m) = k$, so q is a morphism in $\mathbf{XAlg_\Sigma}$.
Thus $\mathbf{XAlg_\Sigma}$ have equalisers.

(f) coequalisers:

- $\mathbf{Alg_\Sigma}$: Coequalisers exist. Let $g, h : A \to B$ be parallel Σ-homomorphisms from $\mathcal{A}$ to $\mathcal{B}$. Define $R := \{(g(a), h(a)) | a \in A\} \subseteq B \times B$. Let Δ be the smallest congruence containing R, i.e. $\Delta := \bigcap\{\theta \subseteq B \times B | \equiv \text{ is a congruence and } R \subseteq \theta\}$. Let $q : B \to B/\Delta$ be the quotient map given by $q(b) = [b]_\Delta$. We turn B/Δ into Σ-algebra by defining $f^{\mathcal{B}/\Delta}([b_1], ..., [b_n]) := [f^{\mathcal{B}}(b_1, ..., b_n)]$ for all $f \in \Sigma$ with arity n. This is well-defined because Δ is a congruence. It follows that q is a Σ-homomorphism. Since $(g(a), h(a)) \in \Delta$ for all $a \in A$, we have $q \circ g = q \circ h$.
  Now, suppose $d : B \to D$ is a homomorphism from $\mathcal{B}$ to $\mathcal{D}$ with $d \circ g = d \circ h$. Consider $\hat{K} := \{(b, b') | d(b) = d(b')\}$. Then $\hat{K}$ is a congruence on $\mathcal{B}$ since d preserves all operations. Because $d(g(a)) = d(h(a))$ for all $a \in A$, we have $R \subseteq \hat{K}$. By minimality of Δ, $\Delta \subseteq \hat{K}$. Define $m : B/\Delta \to D$ by $m([b]_\Delta) := d(b)$. This is well-defined because if $(b, b') \in \Delta$, then $d(b) = d(b')$, so the choice of an equivalence class representative does not matter. Thus $m(q(b)) = d(b)$ for all $b \in B$, so $m \circ q = d$.
  To see that uniqueness holds, let $m' : B/\Delta \to D$ satisfy $m' \circ q = d$. Then for any class $[b]_\Delta$, $m([b]_\Delta) = m(q(b)) = d(b) = m'(q(b)) = m'([b]_\Delta)$, so $m = m'$.
  It remains to show that m is a Σ-homomorphism. If $f \in \Sigma$ has arity 0, then $m(f^{\mathcal{B}/\Delta}) = m([f^{\mathcal{B}}]) = f^{\mathcal{D}}$.
  If $f \in \Sigma$ has arity $n > 0$, then

  $$m(f^{\mathcal{B}/\Delta}([b_1], ..., [b_n])) = m([f^{\mathcal{B}}(b_1, ..., b_n)]) = d(f^{\mathcal{B}}(b_1, ..., b_n)) = f^{\mathcal{D}}(d(b_1), ..., d(b_n))$$

  which in turn equals to $f^{\mathcal{D}}(m([b_1]), ..., m([b_n]))$.
  This concludes the construction of $q : B \to B/\Delta$ and B/Δ as coequaliser of g, h.

- $\mathbf{KAlg_\Sigma}$:
  Coequalisers do not exist for all signatures, and again counterexample can be built with $\Sigma = \{u\}$ with $ar(u) = 1$. Let $\mathcal{B}$ be a Σ-algebra with domain $\{0, 1, 2, 3\}$ and $u^{\mathcal{B}}(0) = 1,\ u^{\mathcal{B}}(1) = 0,\ u^{\mathcal{B}}(2) = 3,\ u^{\mathcal{B}}(3) = 2$. Then $\mathcal{B}$ is an object in $\mathbf{KAlg_\Sigma}$ since every element of B is a kixpoint.

Now consider a pair of parallel arrows in $\mathbf{KAlg_\Sigma}$, $1_B, u^{\mathcal{B}} : B \to B$. Both are Σ-homomorphisms. Assume a coequaliser $q : B \to Q$ exists in $\mathbf{KAlg_\Sigma}$. Then $q = q \circ 1_B = q \circ u^{\mathcal{B}}$ so for all $x \in B$

$$q(x) = q(u^{\mathcal{B}}(x)) = u^{\mathcal{Q}}(q(x)) \quad (q \text{ is a homomorphism}).$$

Hence $u^{\mathcal{Q}} \circ q = 1_Q \circ q$. But every coequaliser is an epimorphism (statement dual to the one known from lectures about regular monos), so from $u^{\mathcal{Q}} \circ q = 1_Q \circ q$ we have $u^{\mathcal{Q}} = 1_Q$. Now $\mathcal{Q}$ is an object in $\mathbf{KAlg_\Sigma}$ and $u^{\mathcal{Q}} = 1_Q$ forces Q to be a singleton: if $|Q| > 1$, then no element k can satisfy kixpoint condition. Thus $\mathcal{Q}$ is a one-element algebra and q is a constant map.
Now consider $D = \{a, b, c\}$ with $u^{\mathcal{D}}(a) = a,\ u^{\mathcal{D}}(b) = b, u^{\mathcal{D}}(c) = a$. Then c is a kixpoint of $\mathcal{D}$, so $\mathcal{D}$ is an object in $\mathbf{KAlg_\Sigma}$. Define $h : B \to D$ by $h(0) = h(1) = a,\ h(2) = h(3) = b$. This h is a homomorphism and satisfies $h = h \circ u^{\mathcal{B}}$, but it is not constant. Hence it cannot factor through the constant map q to a singleton, which is a contradiction.
Thus $\mathbf{KAlg_\Sigma}$ does not have coequalisers for every Σ.

- $\mathbf{XAlg_\Sigma}$:
Coequalisers exist. This is arguably the most involved construction so far, hence I will split it into series of steps.
Step 1.
Take a pair of parallel arrows in $\mathbf{XAlg_\Sigma}$ from $(\mathcal{A}, k)$ to $(\mathcal{B}, l)$ given by $h_0, h_1 : A \to B$. Importantly, h_0, h_1 are also kixpoint preserving Σ-homomorphisms in $\mathbf{Alg_\Sigma}$. Let $q_0 : B \to Q_0$ along with Σ-algebra $\mathcal{Q}_0$ be their coequaliser in $\mathbf{Alg_\Sigma}$. Define $m_0 := q_0(l) \in Q_0$.
At this point $(\mathcal{Q}_0, m_0)$ is a pointed algebra, but we are not guaranteed that m_0 is a kixpoint. We now construct a further quotient $\pi : Q_0 \to Q$ which makes the image of m_0 into a kixpoint.
Step 2.
Let θ be a congruence on Q_0. Consider the quotient algebra $\mathcal{Q}_0/\theta$ and the element $[m_0]_\theta$. Applying the definition of kixpoint we get the following:
$[m_0]_\theta$ is a kixpoint in $\mathcal{Q}_0/\theta$ iff for every $f \in \Sigma$ of arity n,

$$f^{\mathcal{Q}_0}(m_0, ..., m_0) \equiv_\theta m_0 \implies \forall x_1, ..., x_n \in Q_0. f^{\mathcal{Q}_0}(x_1, ..., x_n) \equiv_\theta m_0$$

Let Θ be the set of all congruences θ on Q_0 satisfying the condition above. Observe that Θ is non-empty since it contains the total congruence $Q_0 \times Q_0$ which trivially satisfies the condition (it yields a one element algebra where every operation is forced to be constant). Define $\theta_* = \bigcap_{\theta \in \Theta} \theta$. We know that θ_* is a congruence. Also, we get that $\theta_* \in \Theta$: suppose that $f^{\mathcal{Q}_0}(m_0, ..., m_0) \equiv_{\theta_*} m_0$. Then for every $\theta \in \Theta$ $f^{\mathcal{Q}_0}(m_0, ..., m_0) \equiv_\theta m_0$, so by the definition of Θ, $\forall x_1, ..., x_n \in Q_0. f^{\mathcal{Q}_0}(x_1, ..., x_n) \equiv_\theta m_0$. Thus it holds in the intersection θ_*, so it satisfies the kixpoint condition.
Step 3.
We are finally ready to define the candidate coequaliser. Let $\pi : Q_0 \to Q$ be the quotient map, where $Q := Q_0/\theta_*$. Define $m := \pi(m_0)$. It follows that m is a kixpoint in $\mathcal{Q}$. Hence $(\mathcal{Q}, m)$ is an object in $\mathbf{XAlg_\Sigma}$.
Now define $q := \pi \circ q_0 : B \to Q$. Crucially, $q(l) = (\pi \circ q_0)(l) = \pi(m_0) = m$, so q is a morphism in $\mathbf{XAlg_\Sigma}$. Moreover,

$$qh_0 = \pi q_0 h_0 = \pi q_0 h_1 = qh_1,$$

so q indeed coequalises h_0 and h_1 in $\mathbf{XAlg_\Sigma}$.

Step 4.
It remains to show that the universal property of coequalisers holds.
Suppose that $d : B \to D$ is a homomorphism from $(\mathcal{B}, l)$ to $(\mathcal{D}, n)$ s.t. $dh_0 = dh_1$. Again, we turn to $\mathbf{Alg_\Sigma}$ and we view d as a homomorphism there. Since q_0 is a coequaliser in $\mathbf{Alg_\Sigma}$, there exists a unique homomorphism $d' : Q_0 \to D$ s.t. $d = d' \circ q_0$. Furthermore, $d'(m_0) = d'(q_0(l)) = n$. Let $\theta' := \{(x, y) \in Q_0 \times Q_0 | d'(x) = d'(y)\}$.
Then $\theta' \in \Theta$ for the following reason: Fix $f \in \Sigma$ of arity r. Assume that $f^{\mathcal{Q}_0}(m_0, ..., m_0) \equiv_{\theta'} m_0$, i.e. $d'(f^{\mathcal{Q}_0}(m_0, ..., m_0)) = d'(m_0) = n$. Since d' is a homomorphism, the LHS is equal to $f^{\mathcal{D}}(d'(m_0), ..., d'(m_0)) = f^{\mathcal{D}}(n, ..., n)$. By assumption n is a kixpoint of $\mathcal{D}$, so by the kixpoint condition it follows that $f^{\mathcal{D}}$ is a constant function. Now take arbitrary $x_1, ..., x_r \in Q_0$. Again by homomorphism property, $d'(f^{\mathcal{Q}_0}(x_1, ..., x_r)) = f^{\mathcal{D}}(d'(x_1), ..., d'(x_r)) = n = d'(m_0)$. Thus

$$\forall x_1, ..., x_r \in Q_0 . f^{\mathcal{Q}_0}(x_1, ..., x_r) \equiv_{\theta'} m_0.$$

Hence $\theta' \in \Theta$.
Observe that $\theta_* \subseteq \theta'$. By definition of θ', if $x \equiv_{\theta'} y$, then $d'(x) = d'(y)$, so d' is constant on θ_* equivalence classes. It follows that we obtain a unique homomorphism $u : Q \to D$ s.t. $u\pi = d'$. Moreover, $u(m) = u(\pi(m_0)) = d'(m_0) = n$, so u is a morphism from $(\mathcal{Q}, m)$ to $(\mathcal{D}, n)$ in $\mathbf{XAlg_\Sigma}$.
Finally, $uq = u\pi q_0 = d'q_0 = d$.
Suppose u' from $(\mathcal{Q}, m)$ to $(\mathcal{D}, n)$ also satisfies $u'q = d$. We know that $q_0 : B \to Q_0$ is a quotient map hence it is surjective, and the same applies to $\pi : Q_0 \to Q$. Therefore $q = \pi \circ q_0$ is also surjective. For any $z \in Q$ pick $b \in B$ with $q(b) = z$. Then

$$u(z) = u(q(b)) = d(b) = u'(q(b)) = u'(z),$$

so $u = u'$ which concludes the construction.

(g) (i)

- Let $U : \mathbf{XAlg}_{\mathbf{\Sigma}} \to \mathbf{Alg}_{\mathbf{\Sigma}}$ be a forgetful functor. In this case a left adjoint exists for every Σ. Let $\mathcal{T}_x$ be the term algebra on $\{x\}$. The category $\mathbf{Alg}_{\mathbf{\Sigma}}$ has coproducts, so for any $\mathcal{A}$ we can form the coproduct $\mathcal{A} + \mathcal{T}_x$ with homomorhisms ι_1 from $\mathcal{A}$, and ι_2 from $\mathcal{T}_x$.
Define a candidate functor $L : \mathbf{Alg}_{\mathbf{\Sigma}} \to \mathbf{XAlg}_{\mathbf{\Sigma}}$ by

$$L(\mathcal{A}) := (\mathcal{A} + \mathcal{T}_x, k_{\mathcal{A}}), \quad \text{where } k_{\mathcal{A}} := \iota_2(x)$$

and on morphisms $u : A \to B$ by $L(u) := u + 1_{\mathcal{T}_x}$, where $u + 1_{\mathcal{T}_x}$ is a homomorphism from $\mathcal{A} + \mathcal{T}_x$ to $\mathcal{B} + \mathcal{T}_x$ satisfying

$$(u + 1_{\mathcal{T}_x}) \circ \iota_1 = j_1 \circ u, \quad (u + 1_{\mathcal{T}_x}) \circ \iota_2 = j_2.$$

It is not immediately obvious that $k_{\mathcal{A}}$ is a kixpoint, so consider the following: for any function symbol $f \in \Sigma$ of arity $n > 0$, consider the Σ-algebra $\mathcal{E} = (E = \{0, 1\}, (f^{\mathcal{E}})_{f \in \Sigma})$ where every n-ary function symbol is interpreted as a constant 0 function (and each nullary symbol is interpreted as 0 as well). Importantly, there are two particular homomorphisms from $\mathcal{A}$ to $\mathcal{E}$ and $\mathcal{T}_x$ to $\mathcal{E}$ respectively:
the constant map $h : A \to E$ to 0 and $g : T_\Sigma(\{x\}) \to E$ s.t. $g(x) = 1$.
By the coproduct universal property, these induce a homomorphism ϕ from $\mathcal{A} + \mathcal{T}_x$ to $\mathcal{E}$ s.t. $\phi(k_{\mathcal{A}}) = 1$. Then for any $n > 0$,

$$\phi(f^{\mathcal{A}+\mathcal{T}_x}(k_{\mathcal{A}}, ..., k_{\mathcal{A}})) = f^{\mathcal{E}}(1, ..., 1) = 0 \neq 1 = \phi(k_{\mathcal{A}}),$$

hence $f^{\mathcal{A}+\mathcal{T}_x}(k_{\mathcal{A}}, ..., k_{\mathcal{A}}) \neq k_{\mathcal{A}}$. For 0-ary f the kixpoint condition is automatic (since f is constant anyway). Hence $k_{\mathcal{A}}$ is a kixpoint and $L(\mathcal{A})$ is in $\mathbf{XAlg}_{\mathbf{\Sigma}}$.
It remains to show that $L \dashv U$. We aim to show that there exists a unit $\eta : Id_{\mathbf{Alg}_{\mathbf{\Sigma}}} \to UL$ s.t. $L(\mathcal{A})$ with $\eta_{\mathcal{A}}$ is free over $\mathcal{A}$ wrt. U for all $\mathcal{A} \in Ob(\mathbf{Alg}_{\mathbf{\Sigma}})$. Concretely, we want to show that for every $(\mathcal{B}, l) \in Ob(\mathbf{XAlg}_{\mathbf{\Sigma}})$ and every morphism h from $\mathcal{A}$ to $U((\mathcal{B}, l)) = \mathcal{B}$ in $\mathbf{Alg}_{\mathbf{\Sigma}}$, there exists a unique morphism $h^{\#}$ from $L(\mathcal{A})$ to $(\mathcal{B}, l)$ in $\mathbf{XAlg}_{\mathbf{\Sigma}}$ s.t.

$$U(h^{\#}) \circ \eta_{\mathcal{A}} = h.$$

Let $\eta : Id_{\mathbf{Alg}_{\mathbf{\Sigma}}} \to UL$ have components $\eta_{\mathcal{A}} := \iota_1$ from $\mathcal{A}$ to $\mathcal{A} + \mathcal{T}_x$.
Fix $\mathcal{A} \in Ob(\mathbf{Alg}_{\mathbf{\Sigma}}), (\mathcal{B}, l) \in Ob(\mathbf{XAlg}_{\mathbf{\Sigma}})$ and a homomorphism $h : A \to B$.
There is a unique homomorphism $g_l : T_\Sigma(\{x\}) \to B$ with $g_l(x) = l$. By coproduct universal property of $\mathcal{A} + \mathcal{T}_x$ there is a unique homomorphism $[h, g_l]$ from $\mathcal{A} + \mathcal{T}_x$ to $\mathcal{B}$ satisfying

$$[h, g_l] \circ \iota_1 = h, \quad [h, g_l] \circ \iota_2 = g_l.$$

Set $h^{\#} = [h, g_l]$. We must check if $h^{\#}$ is a morphism in $\mathbf{XAlg}_{\mathbf{\Sigma}}$, i.e. if it preserves the distinguished kixpoints: $h^{\#}(k_{\mathcal{A}}) = l$.
We know that $k_{\mathcal{A}} = \iota_2(x)$, so

$$h^{\#}(k_{\mathcal{A}}) = h^{\#}(\iota_2(x)) = (h^{\#} \circ \iota_2)(x) = g_l(x) = l.$$

Finally, the required equation $U(h^{\#}) \circ \eta_{\mathcal{A}} = h$ is just $h^{\#} \circ \iota_1 = h$, which holds by construction.
It remains to show uniqueness of $h^{\#}$. Suppose some h' from $L(\mathcal{A})$ to $(\mathcal{B}, l)$ also satisfies $U(h') \circ \eta_{\mathcal{A}} = h$. Because h' is a morphism in $\mathbf{XAlg}_{\mathbf{\Sigma}}$, it also satisfies $h'(k_{\mathcal{A}}) = l$. Hence

$$(h' \circ \iota_2)(x) = h'(k_{\mathcal{A}}) = l.$$

Hence $h' \circ \iota_2 : T_\Sigma(\{x\}) \to B$ is a homomorphism sending x to l, so it is forced to satisfy $h' \circ \iota_2 = g_l$. Now h' is a homomorphism from $\mathcal{A} + \mathcal{T}_x$ to $\mathcal{B}$ with

$$h' \circ \iota_1 = h, \quad h' \circ \iota_2 = g_l.$$

By the uniqueness condition in the coproduct universal property, this forces

$$h' = [h, g_l] = h^\#.$$

Thus $L(\mathcal{A})$ with $\eta_\mathcal{A}$ is free over $\mathcal{A}$ wrt. U, so indeed $L \vdash U$.

- Let $U : \mathbf{XAlg_\Sigma} \to \mathbf{KAlg_\Sigma}$ be a forgetful functor. Since for every $(\mathcal{B}, l) \in Ob(\mathbf{XAlg_\Sigma})$ we have $\mathcal{B} \in Ob(\mathbf{KAlg_\Sigma})$, and since for $\mathcal{A} \in Ob(\mathbf{KAlg_\Sigma})$ the coproduct $\mathcal{A} + \mathcal{T}_x$ has the canonical kixpoint $\iota_2(x)$, the previous construction restricts to a functor $L' : \mathbf{KAlg_\Sigma} \to \mathbf{XAlg_\Sigma}$. For $\mathcal{A} \in Ob(\mathbf{KAlg_\Sigma})$ and $(\mathcal{B}, l) \in Ob(\mathbf{XAlg_\Sigma})$ we have

  $$\mathbf{XAlg_\Sigma}(L'(\mathcal{A}), (\mathcal{B}, l)) \cong \mathbf{KAlg_\Sigma}(\mathcal{A}, \mathcal{B}) = \mathbf{Alg_\Sigma}(\mathcal{A}, \mathcal{B}).$$

  naturally. Thus $L' \vdash U$

- Let $U : \mathbf{XAlg_\Sigma} \to \mathbf{Set}$ be a forgetful functor. The left adjoint exists: it is the functor $L : \mathbf{Set} \to \mathbf{XAlg_\Sigma}$ defined by $L(X) = (\mathcal{T}_{X \sqcup \{*\}}, *)$ on objects. On arrows $u : X \to Y$, $L(u)$ is Σ-homomorphism acting as a substitution of variables along u. Concretely:

  - for $x \in X$, $L(u)(x) = u(x) \in Y \subseteq L(Y)$
  - $L(u)(*) = *$
  - for a 0-ary symbol $c \in \Sigma$, $L(u)(c) = c$
  - where $f \in \Sigma$ has arity $n > 0$ for a term $f(t_1, ..., t_n)$ we have

    $$L(u)(f(t_1, ..., t_n)) = f(L(u)(t_1), ..., L(u)(t_n))$$

  Observe that $*$ is a kixpoint in $\mathcal{T}_{X \sqcup \{*\}}$ as every other variable in the term algebra.
  Define the unit $\eta : Id_\mathbf{Set} \to UL$ by components $\eta_X : X \to ULX$ with $\eta_X(x) = x$. Then for every $(\mathcal{B}, l)$ with a function $g : X \to U((\mathcal{B}, l)) = B$ consider an extension $g_* : X \sqcup \{*\} \to B$ with $g_*(x) = g(x),\ x \in X$ and $g_*(*) = l$. Define $g^\# : L(X) \to (\mathcal{B}, l)$ by the following:

  - $g^\#(a) = g_*(a)$ on variables $a \in X \sqcup \{*\}$
  - for 0-ary $c \in \Sigma$, $g^\#(c) = c^\mathcal{B}$
  - for $f \in \Sigma$ of arity $n > 0$ $g^\#(f(t_1, ..., t_n)) = f^\mathcal{B}(g^\#(t_1), ..., g^\#(t_n))$

  It is a Σ-homomorphism by construction and it preserves the distinguished kixpoint, since $g^\#(*) = g_*(*) = l$. Thus $g^\#$ is a morphism in $\mathbf{XAlg_\Sigma}$.
  For $x \in X$ we have

  $$((Ug^\#) \circ \eta_X)(x) = (Ug^\#)(x) = g_*(x) = g(x).$$

  For uniqueness, suppose h from $(\mathcal{T}_{X \sqcup \{*\}}, *)$ to $(\mathcal{B}, l)$ satisfies $Uh \circ \eta_X = g$.
  Then $h(x) = g(x)$ for all $x \in X$ and $h(*) = l$ as a morphism in $\mathbf{XAlg_\Sigma}$. Since h is a Σ-homomorphism, these values on variables determine the values on all terms. Hence $h = g^\#$. Thus LX with η_X is free on X wrt. U, and $L \vdash U$.

(ii) Let $U : \mathbf{KAlg}_{\mathbf{\Sigma}} \to \mathbf{Set}$ be a forgetful functor. In this case a left adjoint does not exist for every Σ. As a counterexample consider the case of empty signature. Then $\mathbf{KAlg}_{\varnothing} = \mathbf{Set}_{\neq\varnothing}$, and it does not have initial objects. Since **Set** has the initial object and left adjoints preserve colimits, a left adjoint cannot exist.
(iii) Let $U : \mathbf{Alg}_{\mathbf{\Sigma}} \to \mathbf{Set}$ be a forgetful functor. In this case a left adjoint exists:
it is $L : \mathbf{Set} \to \mathbf{Alg}_{\mathbf{\Sigma}}$ given by $L(X) = \mathcal{T}_X$ on objects and on arrows $u : X \to Y$, $L(u)$ is Σ-homomorphism acting as a substitution of variables along u. Concretely:

- for $x \in X$, $L(u)(x) = u(x) \in Y \subseteq L(Y)$
- for a 0-ary symbol $c \in \Sigma$, $L(u)(c) = c$
- where $f \in \Sigma$ has arity $n > 0$ for a term $f(t_1, ..., t_n)$ we have

$$L(u)(f(t_1, ..., t_n)) = f(L(u)(t_1), ..., L(u)(t_n))$$

Define the unit $\eta : Id_{\mathbf{Set}} \to UL$ by components given by $\eta_X : X \to U(\mathcal{T}_X)$ with $\eta_X(x) = x$ (where RHS is to be viewed as the variable term in $T_\Sigma(X)$).
Now it remains to show that $L(X)$ with η_X is free over X wrt. U.
Given arbitrary Σ-algebra $\mathcal{A}$ with a function $g : X \to U(\mathcal{A})$ define $g^\# : L(X) \to \mathcal{A}$:

- $g^\#(x) = g(x), \quad x \in X$
- for 0-ary $c \in \Sigma$, $g^\#(c) = c^{\mathcal{A}}$
- for $f \in \Sigma$ of arity $n > 0$ $g^\#(f(t_1, ..., t_n)) = f^{\mathcal{A}}(g^\#(t_1), ..., g^\#(t_n))$

This is a homomorphism by construction and it satisfies $U(g^\#) \circ \eta_X = g$ (by the first point). Uniqueness of $g^\#$ follows immediately, because if $h : T_\Sigma(X) \to A$ is a Σ-homomorphism, it must agree with $g^\#$ on $x \in X$, hence on all terms. Thus $h = g^\#$.
It concludes the argument that $L \vdash U$.
(iv) The inclusion functor $I : \mathbf{KAlg}_{\mathbf{\Sigma}} \to \mathbf{Alg}_{\mathbf{\Sigma}}$:
There is no left adjoint for the empty signature. As before, in this case $\mathbf{KAlg}_{\varnothing} \cong \mathbf{Set}_{\neq\varnothing}$ and $\mathbf{Alg}_{\varnothing} \cong \mathbf{Set}$. Suppose a left adjoint L to I exists. Since left adjoints preserve colimits, $L(\varnothing)$ is initial in $\mathbf{Set}_{\neq\varnothing}$. But there are no intial objects in $\mathbf{Set}_{\neq\varnothing}$, a contradiction. Hence a left adjoint cannot exist for $\Sigma = \varnothing$.

References

[1] S. N. Burris and H. P. Sankappanavar, *A Course in Universal Algebra*, The Millennium Edition — 2012 Update, online PDF, ISBN 978-0-9880552-0-9. Available at `https://www.math.uwaterloo.ca/~snburris/htdocs/UALG/univ-algebra2012.pdf`